01_Ansible Lab logs

Setup:

- Ansible controller: Ubuntu 24.04.3 LTS
(enp0s8: 192.168.0.1/24)
(enp0s9: dhcp4: true)
Additionally works as client

- DHCP server: Ubuntu 24.04.3 LTS
(enp0s8: 192.168.0.2/24)
(enp0s9: 192.168.100.1/24)

Networks (for both machines):
- enp0s3 – NAT (Access to the internet)

- enp0s8 - Internal network (For configurations - static IPs)

- enp0s9 - Internal network (Where DHCP server will work and we will see the results of setup)
Additionaly:

- SSH access to the DHCP server currently relies on password-based authentication. **For production environments, public key authentication should be used** instead to reduce the attack surface and to simplify secure automation and configuration management.

Task:

1) install Ansible on the Ansible controller server

2) enable SSH on the DHCP server

3) configure static IPs on enp0s8

4) make playbook on Ansible controller to setup DHCP server on enp0s9

Steps taken:

--- Ansible controller configuration ---

1) sudo apt update
   sudo apt install ansible -y

2) Building structure in user directory:
   ansible-lab:
    --- dhcp-setup:
        --- inventory
        --- dhcp.yml

   or

   mkdir ansible-lab
   cd ansible-lab
   mkdir dhcp-setup
   cd dhcp-setup
   touch inventory
   touch dhcp.yml

3) sudo nano inventory and adding lines:

   [dhcp_server]
   192.168.0.2 ansible_user=administrator

   --- Where 192.168.0.2 DHCP server IP on enp0s8 and ansible_user=administrator is user on DHCP server

4) ansible -i inventory dhcp_server -m ping --ask-pass --ask-become-pass

--- Testing the connection , made sure ssh is working **on DHCP server with:**

sudo apt install ssh

sudo systemctl start ssh

--- Because we are using passwords for login, Ansible controller must have installed sshpass:

sudo apt install sshpass

--- Received answer "pong", pinged DHCP server successfully using Ansible

```
administrator@ubungu:~/ansible-lab/dhcp-setup$ ansible -i inventory dhcp_server -m ping --ask-pass --ask-become-pass
SSH password:
BECOME password[defaults to SSH password]:
192.168.0.2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
administrator@ubungu:~/ansible-lab/dhcp-setup$
```

5) sudo nano dhcp.yml

- now setting up execution file:

```
  GNU nano 7.2                                          ansible-lab/dhcp-setup/dhcp.yml
---
- name: Automated DHCP installation
  hosts: dhcp_server
  become: yes

  vars:
    interface: enp0s9
    network: 192.168.100.0
    netmask: 255.255.255.0
    range_start: 192.168.100.1
    range_end: 192.168.100.254
    gateway: 192.168.100.1
    dns: 8.8.8.8

  tasks:
    - name: Install DHCP server
      apt:
        name: isc-dhcp-server
        state: present
        update_cache: yes

    - name: Configure default network interface (where dhcp will work)
      lineinfile:
        path: /etc/default/isc-dhcp-server
        regexp: '^INTERFACESv4='
        line: "INTERFACESv4=\"{{ interface }}\""

    - name: Change DHCP configuration
      copy:
        dest: /etc/dhcp/dhcpd.conf
        content: |
          default-lease-time 600;
          max-lease-time 7200;
          authoritative;

          subnet {{ network }} netmask {{ netmask }} {
            range {{ range_start }} {{ range_end }};
            option routers {{ gateway }};
            option domain-name-servers {{ dns }};
          }

    - name: Restart DHCP service
      service:
        name: isc-dhcp-server
        state: restarted
        enabled: yes
```

6) ansible-playbook -i inventory dhcp.yml --ask-pass --ask-become-pass

- Seems to work just fine , DHCP server was set up

```
administrator@ubungu:~/ansible-lab/dhcp-setup$ ansible-playbook -i inventory dhcp.yml

PLAY [Automated DHCP installation] *********************************************************************

TASK [Gathering Facts] *********************************************************************************
fatal: [192.168.0.2]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host via ssh: administrator@1
password).", "unreachable": true}

PLAY RECAP *********************************************************************************************
192.168.0.2                : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescued=0    ignored=0

administrator@ubungu:~/ansible-lab/dhcp-setup$ ansible-playbook -i inventory dhcp.yml --ask-pass --ask-become-pass
SSH password:
BECOME password[defaults to SSH password]:

PLAY [Automated DHCP installation] *********************************************************************

TASK [Gathering Facts] *********************************************************************************
ok: [192.168.0.2]

TASK [Install DHCP server] *****************************************************************************
ok: [192.168.0.2]

TASK [Configure default network interface (where dhcp will work)] **************************************
changed: [192.168.0.2]

TASK [Change DHCP configuration] ***********************************************************************
changed: [192.168.0.2]

TASK [Restart DHCP service] ****************************************************************************
changed: [192.168.0.2]

PLAY RECAP *********************************************************************************************
192.168.0.2                : ok=5    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

administrator@ubungu:~/ansible-lab/dhcp-setup$ _
```

7) ip a

- Checking if DHCP server started working and provided Ansible controller IP address. Ansible controller has received IP address from DHCP server on enp0s9 , everything works as planned.

```
administrator@ubungu:~/ansible-lab/dhcp-setup$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:ac:ab brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
       valid_lft 85769sec preferred_lft 85769sec
    inet6 fd17:625c:f037:2:a00:27ff:fe55:acab/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86262sec preferred_lft 14262sec
    inet6 fe80::a00:27ff:fe55:acab/64 scope link
       valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d5:b6:4b brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.1/24 brd 192.168.0.255 scope global enp0s8
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fed5:b64b/64 scope link
       valid_lft forever preferred_lft forever
4: enp0s9: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ba:ce:28 brd ff:ff:ff:ff:ff:ff
    inet 192.168.100.2/24 metric 100 brd 192.168.100.255 scope global dynamic enp0s9
       valid_lft 419sec preferred_lft 419sec
    inet6 fe80::a00:27ff:feba:ce28/64 scope link
       valid_lft forever preferred_lft forever
administrator@ubungu:~/ansible-lab/dhcp-setup$
```